

Week 3: Core Features & Troubleshooting Report

A Technical Analysis of Event-Handling, Reactive State Synchronization, Rigorous Form Validation, and Defect Resolution on Physical Android Hardware.

Project: FitPulse (com.fitpulse.app)	Version: 2.0.0 (Week 3 Interactive Release)
Framework: Flutter 3.29.0 / Dart 3.7.0	Target Device: TECNO LH8n (Android 14, 1080×2460)
Track: Mobile Application Engineering	Static Analysis: 0 Errors, 0 Warnings (flutter analyze)

1. Executive Summary & Week 3 Objectives

The primary mandate for Week 3 of the Mobile App Engineering Internship is the transition from a static User Interface prototype into an interactive, event-driven, production-grade mobile application. This phase demands the implementation of at least three core interactive capabilities—specifically rigorous multi-input form validations, cross-feature event handling, and dynamic real-time state recalculation—complemented by comprehensive debugging, hardware verification, and formal documentation of all resolved anomalies.

To elevate FitPulse far beyond conventional fitness apps and create a truly premium user experience, we implemented **5 Standout Flagship Features** designed to maximize engagement, retention, and download velocity. The application was thoroughly verified on physical hardware (TECNO LH8n), subjected to zero-lint static analysis, and stress-tested against boundary conditions.

2. Top 5 Flagship Interactive Features Implemented

- Feature 1: Interactive Set & Volume Logger with Real-Time Epley 1RM Calculation**

Athletes can dynamically log workout sets with strict field validation (Weight: 1-600kg, Reps: 1-100, RPE: 1.0-10.0). Features an instantaneous reactive preview of theoretical 1-Rep Max using the Epley formula: $W * (1 + R/30)$. Completing a set automatically activates an intelligent countdown rest timer bar (with +30s increment and Skip controls) and dynamically updates global tonnage across the entire application.

- Feature 2: Dynamic Habit Forge with Strict Modal Validation & Category Analytics**

A customizable habit creation suite with strict title length validation, target goal metrics, and categorized tagging (Fitness, Hydration, Nutrition, Mindset, Wellness). Includes interactive checkbox event handlers, dynamic streak counters, and real-time consistency score recalculation with visual feedback.

• **Feature 3: Macro & Caloric Fast-Logger with Instant Hydration Tracking**

An interactive nutritional modal enabling athletes to input calories and macronutrient splits (protein, carbs, fats) with validation against unrealistic dietary thresholds. Paired with one-tap quick-hydration buttons (+250ml / +500ml) that dynamically recalculate daily caloric rings and hydration progress indicators.

• **Feature 4: Scientific 1-Rep Max Multi-Formula Estimator & Dynamic PR Tracker**

A strength tool comparing three distinct exercise science formulas: Epley, Brzycki ($W * 36 / (37 - R)$), and Lombardi ($W * R^{0.10}$), alongside tailored training intensity zones (95% Power, 85% Strength, 75% Hypertrophy). One-tap 'Save as PR' updates the athlete's personal records live across tabs without requiring screen reloading.

• **Feature 5: Algorithmic Athlete Readiness & Recovery Check-in**

A physiological recovery algorithm evaluating sleep duration (hours), muscle soreness (1-10 scale), energy levels (1-10), and resting heart rate (bpm). Computes an overall Readiness Index (0-100%) and provides contextual coaching recommendations for workout volume auto-regulation.

3. Reactive State Architecture & Event Handling

FitPulse employs a centralized reactive state model built on Flutter's ChangeNotifier and ListenableBuilder primitives. The singleton FitPulseState.instance serves as the single source of truth, ensuring that events in one feature immediately reflect across all other screens without coupling UI widgets.

Layer	Components & Responsibilities
Presentation Layer	Feature Screens (Dashboard, WorkoutTracker, HabitLog, Analytics, Profile). Interacts via ListenableBuilder for zero-overhead local rebuilds.
State & Logic Layer	FitPulseState: Central store managing sets, timers, habits, PRs, readiness metrics, and nutritional logs with atomic mutate-and-notify methods.
Domain Models	Immutable data definitions: WorkoutSet (auto-calculates volume and Epley 1RM), HabitItem, PersonalRecord, ReadinessMetrics.

4. Diagnostic Methodologies & Debugging Tools Utilized

To satisfy the rigorous debugging evaluation criteria, systematic diagnostic practices were deployed throughout the Week 3 sprint:

- Dart Static Analysis (flutter analyze):** Enforced zero compile-time issues, eradicated lint warnings, and surfaced type contract discrepancies before runtime.
- Flutter DevTools & Layout Inspector:** Leveraged to inspect widget constraint trees, monitor render flex dimensions, and diagnose micro-overflows.
- Android Debug Bridge (ADB Logcat):** Monitored live system output (adb logcat -s flutter), tracking async timers, memory leaks, and activity lifecycle states.
- Physical Hardware Auditing (TECNO LH8n):** Direct on-device testing captured font-scaling variance, touch-target responsiveness, and physical pixel boundary behaviors under actual Android 14 conditions.

5. Detailed Bug Log & Root Cause Analysis (RCA)

During iterative hardware testing and feature verification, four critical technical bugs and layout defects were identified, diagnosed, and permanently resolved. The table below details each defect:

Bug ID & Severity	Defect Description & Root Cause	Diagnostic Tool & Resolution
BUG-01 MEDIUM Layout Overflow	RenderFlex Right Overflowed by 8.7 Pixels: On the Dashboard, the quick action row containing '+250ml Water' and 'Log Meal / Macros' rendered a yellow/black stripe on physical device. Root Cause: Fixed container horizontal padding combined with device font scaling exceeded screen width in an unconstrained Row.	Tool: Flutter DevTools & Physical Screenshot. Fix: Wrapped label in Flexible with TextOverflow.ellipsis and adjusted horizontal padding from 12px to 8px.
BUG-02 HIGH UI Rendering	PR Card Horizontal Compression (Vertical Letter Wrap): In AnalyticsScreen, long record strings (e.g. '100.0 kg × 3 reps (110.0 kg 1RM)') consumed excessive horizontal space, crushing the adjacent Expanded exercise title into a 1-character vertical column ('B a r b e l l...').	Tool: Physical Screen Capture verification. Fix: Completely refactored _buildPrCard into a 2-tier card layout separating the exercise title badge from the secondary estimated 1RM chip.
BUG-03 HIGH Build Pipeline	Duplicate Android Application Label Manifest Conflict: During Gradle assembleDebug, build failed with manifest merger error regarding android:label declared simultaneously in merged libraries and root manifest.	Tool: Gradle build console trace. Fix: Synchronized package declaration across build.gradle and AndroidManifest.xml to com.fitpulse.app, resolving duplicate attributes.
BUG-04 MEDIUM State & Memory	Uncancelled Timer Leakage Across Tab Switching: Repeated set completion created overlapping Timer.periodic instances, causing the rest timer to tick at double and triple rates.	Tool: ADB Logcat & console output. Fix: Encapsulated timers inside FitPulseState singleton with defensive cancellation guards before re-spawning and clean dispose() cancellation.
BUG-05 LOW API Deprecation	Deprecated Flutter 3.29 Theme & Color Methods: Compiler warnings regarding deprecated .withOpacity() and legacy ThemeData properties.	Tool: flutter analyze. Fix: Migrated all color instances to modern .withValues(alpha: ...) and updated theme tokens to Material 3 standards.

6. Verification & Quality Assurance Results

Verification Criterion	Expected Outcome	Status
Form Submission & Validation	All modals reject empty/malformed inputs with red error labels; accept valid data cleanly.	PASS (100%)
Cross-Screen Dynamic Updates	Logging sets on Track tab immediately updates Volume & PRs on Stats tab without refresh.	PASS (100%)
Scientific 1RM Computation	Epley, Brzycki, and Lombardi formulas compute within 0.1kg tolerance across 1-30 reps.	PASS (100%)
Static Analysis Gate	flutter analyze returns 'No issues found!' (0 errors, 0 warnings, 0 lints).	PASS (0 Issues)
Physical Hardware Execution	Zero crashes, smooth 60fps rendering, responsive touch handling on TECNO LH8n.	PASS (Verified)

7. Deliverables & Submission Package

All required Week 3 deliverables have been compiled and verified:

- **1. Clean Source Code Archive:** Compressed project archive Week3_Core_Features_FitPulse.zip created after flutter clean.
- **2. Troubleshooting & Bug Report:** This official PDF document (Week3_Troubleshooting_and_Bug_Report.pdf) documenting features, RCA, and diagnostics.
- **3. High-Resolution Screenshots:** 17 verified screenshots captured directly from the physical device saved in screenshots/week3/.
- **4. Comprehensive Documentation:** Root README.md updated with architecture specifications, feature logs, and execution guide.

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